How to solve a gas-law exercise, step by step

1 — Read it once

Read the whole exercise to understand the situation.

2 — Read it a second time

Now look for the numbers and the units, and underline them.

3 — Draw the table

Data	Unknown

4 — Write the data

Mark each one as **initial (1)** or **final (2)**: p_1, V_1, T_1... T in **kelvin**!

5 — Write the unknown

What does the exercise ask for? For example, V_2.

6 — What stays constant?

T → **Boyle-Mariotte**
V → **Gay-Lussac**
p → **Charles**

7 — Write the law

Its name and its formula, e.g. $\mathbf{p_1 \cdot V_1 = p_2 \cdot V_2}$

8 — Isolate the unknown

Leave it alone on one side: $\mathbf{V_2 = p_1 \cdot V_1 / p_2}$

9 — Substitute and calculate

Put the numbers **with their units**, operate, and check that the result makes sense.